

Product Backlog: Mumbai–Ahmedabad High-Speed Rail Corridor – Timetable Planning Solution

Epic 1: Project Initiation & Stakeholder Engagement

1.1 Feature: Requirements Elicitation and Analysis

- **User Story:** As a business analyst, I want to gather and clarify all requirements from NHSRCL to ensure the solution meets operational and regulatory needs.
 - Task: Organize and conduct kickoff meetings with NHSRCL stakeholders.
 - Task: Develop a communication plan for all project stakeholders.
 - Task: Review existing plans and international standards.

1.2 Feature: Project Planning and Scheduling

- **User Story:** As a project manager, I want to define objectives, create the schedule, and plan resources so that the project can be executed efficiently and within budget.
 - Task: Create project schedule and milestones.
 - Task: Develop the project budget.
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Epic 2: Analysis & Data Preparation

2.1 Feature: Data Collection and Validation

- **User Story:** As a data engineer, I want to collect and validate geographic, infrastructure, and historical timetable data so that the scheduling models have high-quality input.
 - Task: Gather geographic, infrastructure, and operational data.
 - Task: Integrate data from existing high-speed rail networks as references.
 - Task: Audit and clean collected data for accuracy.

2.2 Feature: Demand and Technological Analysis

- **User Story:** As a system architect, I want to forecast passenger demand and analyze peak hours so that schedules can be optimized for real-world needs.
 - Task: Forecast peak/seasonal demand.
 - Task: Evaluate existing planning tools and software.
 - Task: Identify interfaces with NHSRCL's current IT infrastructure.
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Epic 3: Timetable Solution Design & Development

3.1 Feature: Scheduling System Design

- **User Story:** As a software architect, I want to design an AI/ML-powered timetable generation system that can handle dynamic and automated scheduling.
 - Task: Define scheduling frequency and intervals.
 - Task: Design system architecture (cloud-based, hybrid, modular).
 - Task: Define APIs for integration with weather, passenger info, and signaling systems.

3.2 Feature: Algorithm Development & Simulation

- **User Story:** As a data scientist, I want to develop and test optimization algorithms so that the timetable minimizes delays and maximizes efficiency.
 - Task: Develop multi-criteria optimization algorithms.
 - Task: Simulate timetables to test efficiency and resolve bottlenecks.
 - Task: Adjust timetables based on simulation results.

3.3 Feature: Real-Time Monitoring and Control

- **User Story:** As an operator, I want real-time monitoring of train operations to identify and resolve delays instantly.
 - Task: Implement real-time dashboard with performance monitoring.
 - Task: Enable dynamic adjustments for delays, maintenance, or emergencies.

3.4 Feature: User Access and Security

- **User Story:** As an NHSRCL admin, I want role-based access control and secure authentication for the scheduling platform.
 - Task: Implement RBAC for different user roles.
 - Task: Add multi-factor authentication for critical operations.
 - Task: Set up audit logging and compliance checks.
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Epic 4: Documentation, Training, and Compliance

4.1 Feature: Documentation and Training Materials

- **User Story:** As a project manager, I want to produce clear documentation and training guides so NHSRCL staff can use and maintain the system.
 - Task: Write System Requirements Specification (SRS).
 - Task: Create user manuals and onboarding materials.
 - Task: Conduct staff training sessions.

4.2 Feature: Quality Assurance and Compliance

- **User Story:** As a QA lead, I want to validate deliverables against quality standards to ensure compliance with NHSRCL and international regulations.
 - Task: Plan and conduct quality reviews and audits.

- Task: Document compliance with ISO 9001:2015, UIC standards, and Indian Railways regulations.
 - Task: Implement test automation and monthly data accuracy audits.
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Epic 5: Proposal Preparation, Review, and Submission

5.1 Feature: Proposal Document Development

- **User Story:** As a project manager, I want to prepare a comprehensive, persuasive proposal with clear technical and financial analysis for NHSRCL review.
 - Task: Draft executive summary, scope, methodology, and cost-benefit analysis.
 - Task: Develop visual aids and PowerPoint presentations.

5.2 Feature: Proposal Presentation and Feedback

- **User Story:** As a team lead, I want to present the proposal to NHSRCL, collect feedback, and refine the submission for final approval.
 - Task: Present the proposal to stakeholders.
 - Task: Address stakeholder inquiries and feedback.
 - Task: Update proposal and resubmit as necessary.
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Epic 6: Risk and Issue Management

6.1 Feature: Risk Identification and Tracking

- **User Story:** As a risk manager, I want to maintain a Risk Register and proactively address project threats and opportunities.
 - Task: Identify technical, operational, financial, regulatory, and stakeholder risks.
 - Task: Assess risks using a Probability-Impact Matrix.
 - Task: Track and review risks monthly via JIRA.

6.2 Feature: Risk Mitigation and Reporting

- **User Story:** As a project manager, I want to implement risk responses and report on risk status to minimize impact on project objectives.
 - Task: Develop mitigation plans for high-priority risks.
 - Task: Monitor triggers and implement response actions.
 - Task: Prepare and share risk reports with NHSRCL.
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Epic 7: Project Closure and Lessons Learned

7.1 Feature: Project Review and Closure

- **User Story:** As a project sponsor, I want to formally review the project, gather lessons learned, and ensure all documentation is archived.
 - Task: Summarize key insights and lessons learned.
 - Task: Identify improvements for future projects.
 - Task: Archive all project documentation and code.
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Prioritization

- Initial focus: Requirements elicitation, data collection, and stakeholder communication.
- Subsequent sprints: Technical solution design, algorithm development, and real-time monitoring.
- Parallel activities: Documentation, quality assurance, and risk management.
- Final sprints: Proposal development, feedback incorporation, and project closure.